

Yttrium

39
Y Along with three other elements—ytterbium, erbium, and terbium—yttrium is named after the town of Ytterby, in Sweden. Yttrium has a radioactive form, yttrium-90, which is used to treat some types of cancer.



Laboratory sample of pure yttrium

YAG laser beam

Medical laser

When combined with other elements, yttrium can be used in laser surgery. A YAG (yttrium aluminium garnet) laser, when combined with an element called neodymium, is useful for cosmetic treatment, such as skin laser therapy.



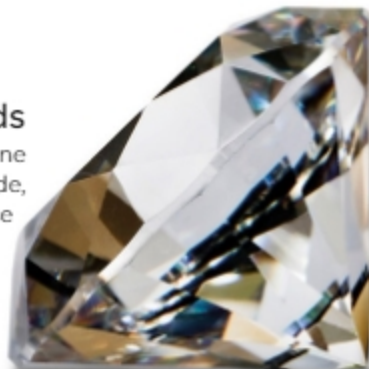
Bar of pure zirconium refined in a laboratory

Zirconium

40
Zr This element gets its name from the Arabic word *zarkûn*, meaning “gold color.” This is because its main mineral source, zircon, has a brownish-golden color. A shiny metal, zirconium is used to make hard-wearing alloys, such as those fitted in the outer layer of fuel rods in nuclear reactors.

Fake diamonds

Cubic zirconia, the crystalline form of zirconium dioxide, can mimic diamonds. These fake diamonds can also appear in various colors—for example, erbium makes pink cubic zirconia.



Molybdenum

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Mo First isolated in 1781 by the Swedish chemist Peter Jacob Hjelm, molybdenum is an essential element in our diet. It is found in foods such as lentils and beans, and it helps clear some toxins in the body. One of its most important industrial uses is in the production of moly-steels—steel alloys with molybdenum added to them to make them harder and more resilient.

Element of war

Both the British and German armies used molybdenum during World War I. While the British used it to protect their tanks, the Germans used it for their heavy artillery.



Chunk of pure molybdenum refined in a laboratory



British World War I Mark IV tank

Steel-molybdenum body forms a protective layer against enemy fire